

# State-of-the-Art Techniques for Fruit Maturity Detection

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**Abstract:** For decades, fruit maturity assessment in the field was challenging for producers, researchers, and food supply agencies. Knowing the maturity stage of the fruit is significant for precision production, harvest, and postharvest management. A prerequisite is to detect and classify fruit of different maturities from the background environment. Recently, deep learning technology has become a widely used method for intelligent fruit detection, due to it having higher accuracy, reliability, and a faster processing speed compared with traditional image-processing methods. At the same time, spectral imaging approaches can predict the maturity stage by acquiring and analyzing the spectral data of fruit samples. These maturity detection methods pay more attention to the species, such as apple, cherry, strawberry, and mango, achieving the mean average precision value of 98.7% in apple fruit. This review provides an overview of the most recent methodologies developed for in-field fruit maturity estimation. The basic principle and representative research output associated with the advantages and disadvantages of these techniques were systematically investigated and analyzed. Challenges, such as environmental factors (illumination condition, occlusion, overlap, etc.), shortage of fruit datasets, calculation, and hardware costs, were discussed. The future research directions in terms of applications and techniques are summarized and demonstrated.

**Keywords:** fruit; maturity detection; spectral analysis; image processing; deep learning

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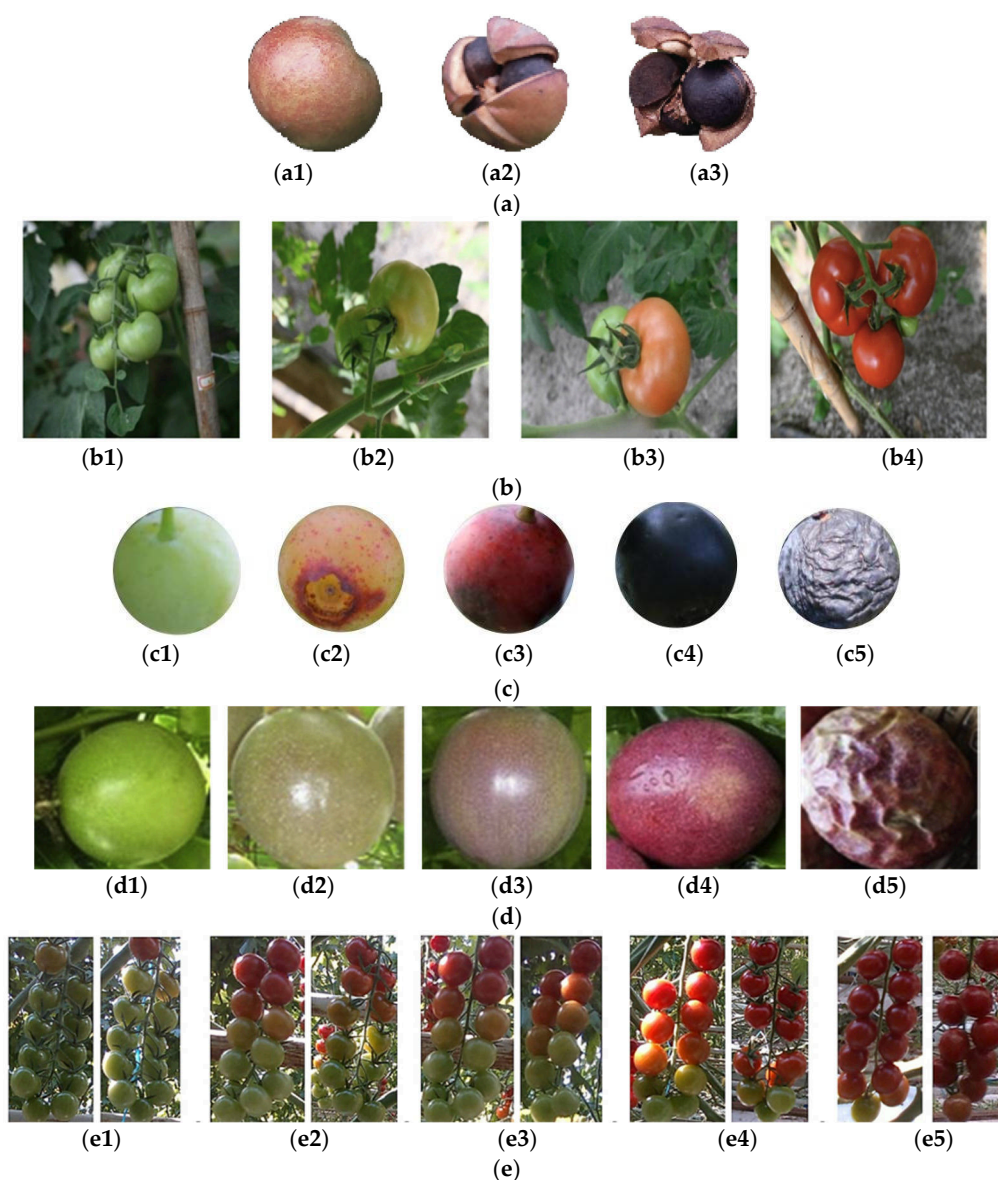
## 1. Introduction

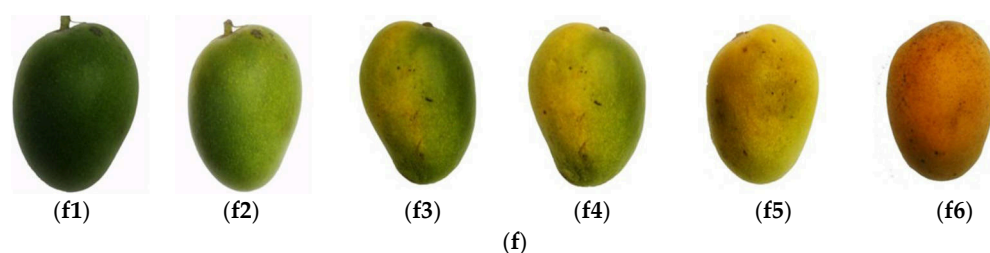
With the rapid growth of the population and acceleration of urbanization, agricultural production of fresh fruit must increase significantly to relieve the pressure on food demand [1]. The principles dictating at which stage of maturity a fruit should be harvested are essential for its subsequent storage, transportation, marketable life, and quality [2]. If fruits are not harvested at the proper stage of maturity, physiological processes occur that permanently change their taste, appearance, and quality. The color, texture, fiber, and consistency of all fruits are greatly affected by the stage of maturity at harvest, as shown in Figure 1. Harvesting too late can result in poor quality due to the development of dietary fiber. However, accurately assessing the maturity of fruits and vegetables has been a repetitive, manual, and labor-intensive process, relying on the subjective judgements of experienced people, which contradicts the increasing labor shortage. The labor cost of fruit harvesting accounts for 30–50% of the total production cost [3,4]. Harvesting fruit is a seasonal task, which requires the growers to find enough labor for timely harvesting. For example, about 30–35 h of skilled labor per acre is required to harvest apple trees [4]. Thus, automatically detecting the maturity of the fruit in the field and accurately locating them are agricultural problems in which spectral and computer vision techniques play fundamental roles [5].

The accuracy, speed, and robustness of object detection and recognition are directly related to the harvesting efficiency, quality, and speed. RGB, spectral, and depth cameras are widely used in harvesting robots to obtain color, texture, and other maturity-related

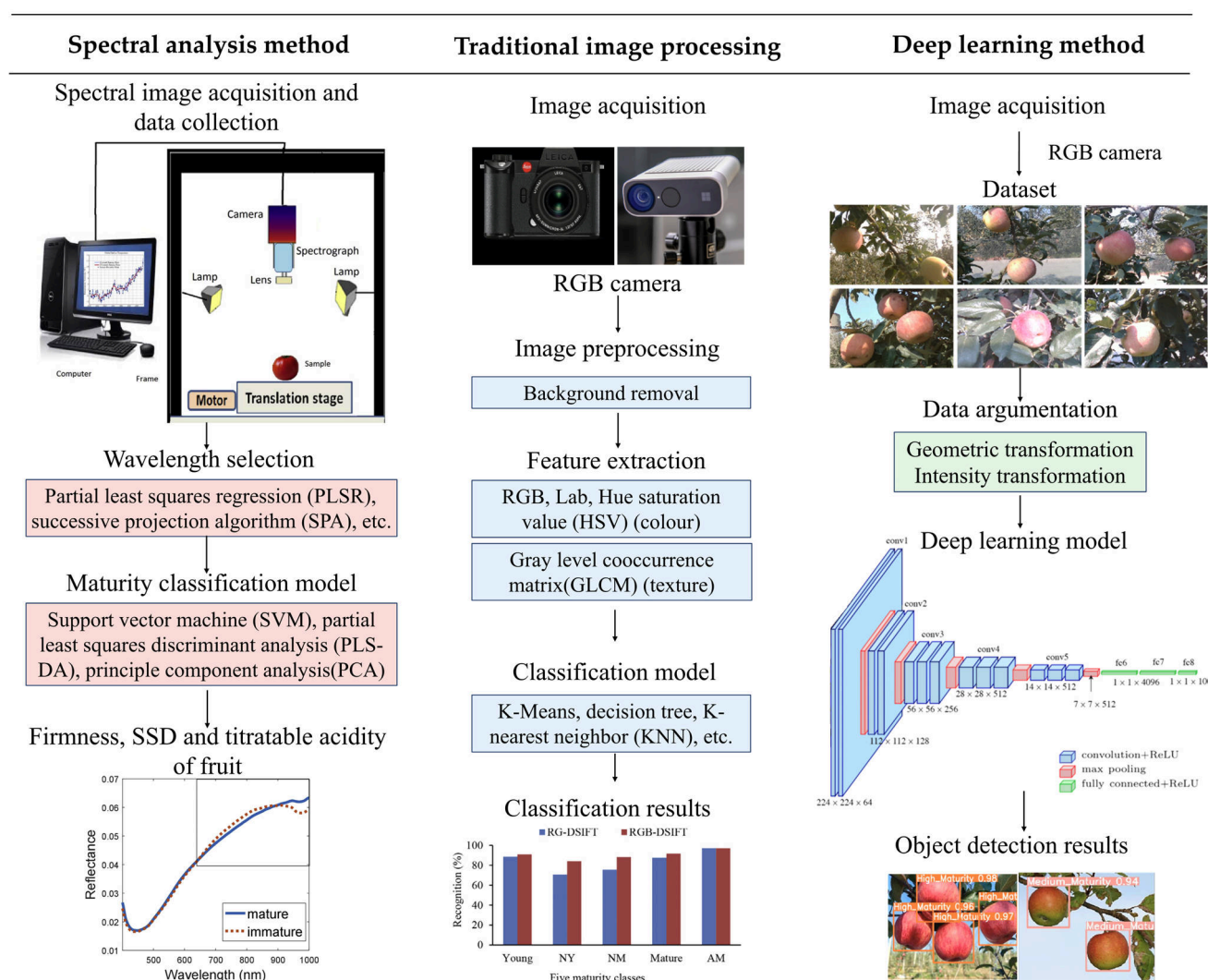
information about fruit in the structured/non-structured environment. The workflow of different methods for fruit maturity detection and recognition is summarized in Figure 2. Many researchers have conducted extensive and in-depth research on fruit maturity detection and classification techniques, which can be subdivided into the following aspects [6,7]:

1. The spectral analysis method includes fruit spectral data acquisition by using Raman imaging (RI), hyperspectral imaging (HSI) or near-infrared (NIR) spectroscopy, data preprocessing, data dimension reduction, wavelength selection, and maturity classification modeling.
2. Computer vision techniques are based on digital image processing, where the color, texture, and shape features are extracted during the fruit growth stages. Then, classifiers based on machine learning, such as support vector machine (SVM), K-means clustering, and decision trees, are adopted for maturity estimation.
3. Deep learning techniques, such as the single-shot multi-box detector (SSD), you only look once (YOLO) model, and Faster region-based convolution neural network (Faster R-CNN) model, are widely used in orchards and greenhouses to achieve high-accuracy and high-robustness real-time detection in complex environments.





**Figure 1.** Examples of fruits of different maturities. (a) *Camellia oleifera* fruit in unripe ((a1) shell without cracks), ripe ((a2) shell with slight cracks), and overripe stages ((a3) rough skin, shell cracks, and seed exposed). (b) Tomato in mature green (b1), breakers (b2), pick (b3), and red (b4) stages. (c) Blueberry in unripe ((c1) green), beginning to ripen ((c2) slightly red), semi-ripe ((c3) red), fully ripe ((c4) purplish black), and overripe stages ((c5) purplish black with texture). (d) Passionfruit in young ((d1) green), near-young ((d2) 20–50% red), near-mature ((d3) 50–80% red), mature ((d4) 90–100% red), and after-mature stages ((d5) fully red with texture). (e) Cherry tomato in 20% red (e1), 40% red (e2), 60% red (e3), 80% red (e4), and 100% red (e5) stages. (f) Mango in dark olive green (f1), olive (f2), apple green (f3), brown (f4), saddle brown (f5), and dark golden (f6) stages.



**Figure 2.** The workflow of different methods for fruit maturity detection and recognition.

The current reviews on fruit ripeness mainly focus on the spectral analysis approaches, in which the machine learning and deep learning models are not included. By critically reviewing the latest research and technological advancement in this field, we

aimed to contribute by (1) systematically summarizing the recent advances in fruit ripeness detection and classification based on spectral analysis, traditional image-processing, and deep learning techniques, (2) scientifically comparing the advantages and disadvantages of various methods, as shown in Tables 1 and 2, and analyzing their innovative practice in the fruit industry, and (3) demonstrating the current challenges affecting detection performance and prospective future potential developments, as summarized in Table 3.

**Table 1.** Advantages and disadvantages of spectral techniques.

| Techniques | Base     | Advantages                                                                                                                          | Disadvantages                                                                                                                                                       |
|------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NIR        | Infrared | It is a non-destructive technique [6].<br>Accurately detects target molecules, e.g., soluble solids content.                        | Requires knowing the concentration of the sample. Fails to provide the spatial distribution of content. Highly affected by the thickness of the fruit sample [8,9]. |
| HSI        | Spectral | Traces the content and concentrations of multiple components in the sample.<br>Provides rich spectral and spatial information [10]. | Has low adaptability of multivariate models and requires high hardware costs.                                                                                       |
| RI         | Spectral | Provides details of vibrations of covalent bonds in the sample.                                                                     | Has weaker Raman scattering effects and interference of biological fluorescence.<br>Requires high hardware costs [11].                                              |

**Table 2.** Advantages and disadvantages of traditional image-processing and deep learning methods.

| Techniques                           | Base          | Advantages                                                              | Disadvantages                                                                           |
|--------------------------------------|---------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Traditional image-processing methods | RGB images    | Convenient to interpret and requires low computational costs.           | Has low accuracy, poor robustness, and a lack of generalization ability.                |
| Deep learning approaches             | Image dataset | Has high accuracy, strong robustness, and great generalization ability. | Requires a large dataset and high computational costs.<br>Has limited interpretability. |

**Table 3.** Challenges and potential improvements for three types of fruit maturity techniques.

| Techniques                          | Challenges                                                                   | Improvements                                                             |
|-------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Spectral analysis method            | Has high sensitivity to illumination and high hardware costs.                | Develop low-cost hardware and introduce artificial intelligence methods. |
| Traditional image-processing method | Low detection accuracy and not robust to environment factors.                | Integrate with deep learning network to improve model accuracy.          |
| Deep learning method                | Lack of datasets, requires high computation costs, and is hard to interpret. | Develop network optimization strategies.                                 |

The present review is structured in six main sections. Section 1 comprises this introduction. Sections 2–4 deal with fruit maturity detection, reviewing the spectral analysis, traditional digital image-processing, and deep learning methods, respectively. In Section 5, the challenges to be faced when applying fruit maturity detection are discussed. Conclusions are proposed in Section 6.

## 2. Fruit Maturity Detection Based on Spectral Analysis

Non-destructive spectroscopic techniques, such as Raman imaging, near-infrared spectroscopy, hyperspectral imaging, nuclear magnetic resonance spectroscopy, etc., have been successfully exploited for rapid detection of ripening and maturity stages of fruits due to their non-destructive nature, environmental friendliness, reliability accuracy, and simplicity [10,12,13]. The advantages and disadvantages of these imaging techniques are

summarized in Table 1. Specifically, Table 4 presents the latest applications of different spectral techniques in fruit ripening and maturity assessment.

**Table 4.** Recent applications of spectral imaging techniques for fruit maturity evaluation.

| Crops             | Attribute                         | Tools | Spectral Range                         | Model                                                       | Performance                              | Advantages                                 | Ref. |
|-------------------|-----------------------------------|-------|----------------------------------------|-------------------------------------------------------------|------------------------------------------|--------------------------------------------|------|
| Apple             | Polysaccharide, pectin            | RI    | 530 nm                                 | K-means                                                     | /                                        | Real-time monitoring                       | [14] |
| Pomegranate       | Tannin                            | RI    | 100–3000 cm <sup>-1</sup>              | Self-modeling mixture analysis (SMA)                        | Ratio of performance deviation (RPD) > 5 | No cutting in sampling                     | [15] |
| Oil palm fruit    | β-carotene, amino acid, etc.      | RI    | 532 nm                                 | K-nearest neighbor (KNN)                                    | Precision: 90.9%                         | Field assessment suitable                  | [16] |
| Peppers           | Maturity index                    | RI    | 400–2300 cm <sup>-1</sup>              | Standard normal variate, principal component analysis (PCA) | /                                        | Portable, streamlined, field-ready process | [17] |
| Durian            | Dry matter content                | NIR   | SWNIR 450–1000 nm<br>LWNIR 860–1750 nm | Linear discriminant analysis (LDA)                          | Precision: 97.28%                        | High accuracy                              | [18] |
| Tomato            | Soluble solids content (SSC)      | NIR   | 650–1000 nm                            | Partial least squares regression (PLSR)                     | Precision: 80.08%                        | High accuracy of SSC                       | [19] |
| Pineapple         | SSC/color                         | NIR   | 200–1100nm                             | PLSR                                                        | Correlation value: 0.97                  | Quality control for mass production        | [20] |
| Mango             | Dry matter and °Brix              | NIR   | 400–1100 nm                            | Multiple linear regression (MLR)                            | Precision: 88.2%                         | Portable, onsite operation                 | [21] |
| Peanut            | Color and maturity                | HSI   | 400–1000 nm                            | Linear mixture model (LMM)                                  | Precision: 95.2%                         | Suitable for protective testing            | [22] |
| Strawberry        | Spectral data and texture feature | HSI   | 380–1030 nm<br>874–1734 nm             | PCA, support vector machine (SVM)                           | Precision: 99.17%                        | High accuracy                              | [23] |
| Strawberry        | SSC                               | HSI   | 380–1030 nm                            | ResNet                                                      | Precision: 84%                           | High accuracy, supports visualization      | [24] |
| Camellia oleifera | Maturity index                    | HSI   | 374.98–1038.8 nm                       | Successive projection algorithm (SPA)                       | Precision: 92.0%                         | Easy to judge harvest timing               | [25] |

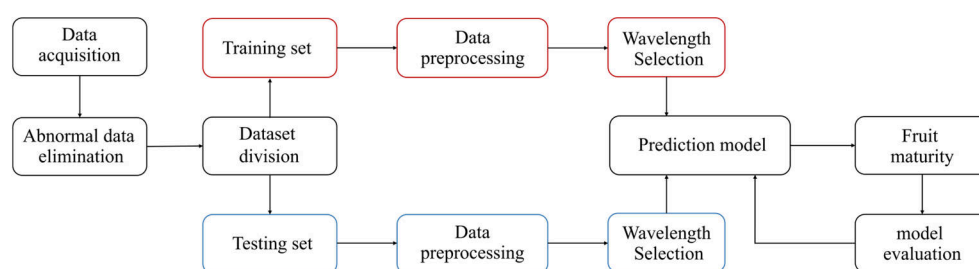
### 2.1. Raman Imaging

Raman imaging integrates Raman spectroscopy and digital imaging to acquire pixels based on spectral and spatial information of samples [14]. In [15], Raman spectroscopy was used to obtain the spectrum in the wavelength range of 100–3000 cm<sup>-1</sup> of the fruit cross-section to detect tannin changes of pomegranate fruit in immature, fairly half-ripe, half-ripe, and fully half-ripe stages. Self-modeling mixture analysis (SMA) was employed to extract component spectra from fruit samples. Three significant Raman peaks corresponding to pure tannin were observed at 650, 1357, and 1590 cm<sup>-1</sup>. Similarly, Azmi et al. [16] developed a ripeness classification algorithm for oil palm fresh fruit utilizing machine learning models and Raman spectral data. The weighted K-nearest neighbor (KNN) and tri-layered neural network classifier achieved an overall accuracy of 90.9% by using four Raman spectral feature components extracted from the peaks as the model predictors. To

determine the ripeness onsite in an automated manner, Legner et al. [17] utilized a non-invasive portable Raman spectrometer combined with the chemometric model to assess the maturity stage for carotenoid-containing fruit, where the change in carotenoids and chlorophylls was recorded during the ripening phase. However, the excitation of the RI system relies on the laser as the energy source. The extensive utilization of point, linear, and global lasers increases the hardware costs. On the other hand, the absence of efficient substrates has limited the application of enhanced RI technology.

## 2.2. NIR Spectroscopic Technology

The maturity of fruits can be characterized by changes in the soluble solids content, titratable acidity, moisture contents, and pH value. The NIR spectrum can be classified into the long NIR spectrum and the short NIR spectrum. The short NIR has a deeper penetration capability beneath the skin that allows it to detect glucose molecules more accurately, compared with the long NIR. NIR spectroscopy has been widely used to measure fruit maturity parameters. Figure 3 shows that this technology is implemented by fruit sampling, NIR spectra acquisition, outliers' elimination, data preprocessing, dataset division, wavelength range selection, and maturity prediction modeling. In addition, chemometric analysis, as a significant aspect of NIR technology, can accurately determine maturity parameters for model calibration. Portable NIR devices have been applied to the fruit maturity detection field in the literature. In [18], due to the thickness of durian fruit, the authors comprehensively utilized short NIR spectra (450–1000 nm) and long NIR spectra (860–1750 nm) to predict four growth stages of durian fruit. The experiment results showed that a combination of rind spectra and stem spectra effectively increased the classification accuracy. Huang et al. [19] used a NIR full-transmittance spectra for cherry maturity evaluation. A prediction model of soluble solids content and a maturity classification model were established by partial least squares regression (PLSR) and extreme learning machine (ELM), respectively. The best results of the total, immature, intermediate, and mature stages of the prediction set achieved 91.31%, 95.38%, 89.23%, and 91.35%. The ELM model presented higher scalability and lower computational complexity by obtaining the minimum training errors. Following this, Semyalo et al. [20] collected pineapple spectral data in the spectral range of 200–1100 nm with an online spectrometer. In addition to the SSC value, the authors evaluated the internal color of the fruit to achieve a correlation coefficient of double cross-validation ( $R_v$ ) of 0.97 and a root mean square error of double cross-validation (RMSEV) of 0.034. Moreover, the idea behind these methods is to predict maturity index values using regression techniques and then apply a hard threshold to the predicted values to estimate the maturity state of fruits. To classify the fruit maturity directly, Ali Shah et al. [21] proposed a novel approach to directly classify on-tree mango fruit maturity based on the reference maturity index, in which a handheld NIR maturity meter was developed to acquire spectral data, preprocess data, reduce data dimensionality, and predict the fruit maturity.



**Figure 3.** The work principles of NIR spectroscopic technology.

However, NIR spectroscopic studies need a set of known sample concentrations for model calibration. Thus, the method's performance relies on the accuracy of the sample presentation. Moreover, the model is required to be calibrated for different fruit varieties,

and this can be addressed by developing a real-time calibration method in the future. On the other hand, the performance of NIR techniques is limited by the thickness of the fruit sample. The problem of spectral distortion occurs when the sample thickness is more than 10 mm. In addition, NIR is appropriate for single-point detection. The uneven distribution of internal components during the ripening stage will cause inaccurate detection.

### 2.3. Hyperspectral Imaging

Hyperspectral imaging incorporates spectroscopy techniques and traditional image-processing or computer vision techniques to acquire spatial and spectral information of samples, in which the spectroscopy techniques provide spectroscopy information, and image-processing methods display spatial data [6]. This technique effectively overcomes the subjectivity and inherent limitations of traditional methods. Zou et al. [22] developed a peanut maturity classification system utilizing the HIS method without removing the exocarp. In the system, a linear mixture model (LLM) and fully constrained least squares (FCLS) classifier were employed for hyperspectral image analysis and maturity classification. Similarly, in [23], the ripeness of the strawberry was determined by analyzing hyperspectral images in the spectral ranges of 441.1–1013.97 nm and 941.46–1578.13 nm with the PCA model. Furthermore, a classification model based on a SVM was built using the optimal wavelength of 441.1–1013.97 nm to obtain the best accuracy of over 85%. In order to develop a non-destructive and automatic detection system, Su et al. [24] integrated a deep learning approach and hyperspectral imaging to determine the maturity and soluble solids content of strawberries. In the proposed method, the preprocessed 1D and 3D hyperspectral image were employed to build 1D ResNet and 3D ResNet, where the results showed that 1D spectral image identified the maturity degree with a small number of samples and lower computation costs. The 3D ResNet, on the other hand, showed a greater advantage when handling large numbers of samples. Numerous studies have shown that the application of deep learning methods in HSI shows great potential to significantly improve the accuracy and robustness of maturity assessment [26]. Furthermore, Yuan et al. [25] recently utilized a hyperspectral imager combined with chemometrics and deep learning approaches to determine the comprehensive maturity index and stages of *Camellia oleifera* fruit in the field. This approach used the successive projection algorithm (SPA) to reduce the dimensions of spectral data and construct a maturity classification model based on partial least squares discriminant analysis (PLS-DA) and fused spectra data to achieve the highest accuracy. Generally, the HIS technique provides a comprehensive way to assess fruit maturity by considering including image and physicochemical information. However, the cost of HIS equipment is relatively high. The process of data acquisition and modeling is very time-consuming, and it has great demand for storage capacity and hardware, which are urgent problems to be solved in the future.

### 3. Fruit Maturity Detection Based on Traditional Image-Processing Techniques

Generally, the process of fruit maturity classification using traditional image-processing methods mainly consists of two steps. Firstly, the maturity-related features, such as color, texture, and geometry, are extracted from fruit color images. Then, the machine-learning-based classifiers are used to determine the ripeness of fruits. Table 5 summarizes the feature extraction methods and classification techniques for fruit ripeness detection and presents the performance of various techniques.

**Table 5.** Fruit maturity detection based on traditional image-processing techniques.

| Crop      | Feature Extraction                                    | Classifiers | Accuracy | Limitations           | Ref  |
|-----------|-------------------------------------------------------|-------------|----------|-----------------------|------|
| Blueberry | Histogram of oriented gradient (HOG) and Lab features | KNN         | 92.06%   | High misjudgment rate | [27] |
| Blueberry | Color feature                                         | SK-means    | 85–98%   | Poor light robustness | [28] |

|          |                                                   |                                 |        |                                                          |      |
|----------|---------------------------------------------------|---------------------------------|--------|----------------------------------------------------------|------|
| Oil palm | Mean and entropy features in color space          | ANN                             | 98.3%  | Complex calculations and long processing times           | [29] |
| Oil palm | Gabor waves, GLCM, and BGLAM texture features     | SVM                             | 93%    | High computational costs                                 | [30] |
| Lycium   | RGB, HSV, and Lab features                        | SVM                             | 100%   | High computational costs and poor generalization ability | [31] |
| Mango    | RGB and HSI features                              | Decision tree                   | 96%    | Sensitive to similar colors                              | [32] |
| Loofah   | HSV and GLCM features                             | Random forest model             | 91.12% | Misdetects small, obscured objects                       | [33] |
| Apricot  | RGB, Lab, and GLCM features                       | Quadratic discriminant analysis | 92.3%  | Not robust to light conditions                           | [34] |
| Mulberry | Color and geometrical features, and GLCM features | Artificial neural network (ANN) | 98.26% | Not robust to shadows and high light                     | [35] |

### 3.1. Fruit Maturity Feature Extraction

In nature, the color, texture, and shape of fruits will change significantly at different maturity stages, which can be used as an important basis to detect and identify fruit maturity. Many researchers have carried out extensive research on fruit maturity detection based on color features (red, green, and blue (RGB) [28,31,32], hue saturation value (HSV), [31,33], Lab [27], and luminance chrominance components (YCbCr) [29]), shape features [35], texture features [33–35], and multiple feature fusion [35].

During ripening, fruits exhibit transition from green to a variety of pigmentations. The techniques based on color features are mainly suitable for the case of fruits with significant color differences at different maturity stages, such as blueberries [27,28], mulberries [35], mangos [32], oil palm fruit [29,30], and *Lycium barbarum* L. [31]. Milm et al. [32] proposed an automatic detection method to classify mangoes into six maturity stages. The RGB and HSI color features of mangoes were extracted from RGB images, and the features with the most information were selected for the mango ripening classification. The detection error of this method is affected by different light intensities. Following this idea, in [31], nine components, including R, G, B, H, S, V, L, a, and b, were extracted from the ripe, half-ripe, and unripe *Lycium barbarum* L. images. Components R, G, and S were identified to be beneficial for extraction of the ripe, half-ripe, and unripe fruit areas. Septiarini et al. [29] proposed a maturity classification model of oil palm fruit by mean and entropy features in YIQ and YCbCr color spaces. The model detected three maturity levels (raw, ripe, and half-ripe) based on the input image. Tu et al. [36] integrated the dense scale invariant features transform (DSIFT) algorithm and locality-constrained linear coding to extract red, green, and blue features in the fruit maturity stages.

Along with color, fruit ripening is often accompanied by changes in texture. The texture features provide another set of indicators for the detection of fruit maturity. Techniques for texture feature extraction include the gray-level cooccurrence matrix (GLCM), Gabor transform, and basic grey-level aura matrix (BGLAM). These techniques are mainly utilized for cases where the textures of fruit are significantly different at different maturity stages. For example, in [30], Gabor waves, GLCM, and BGLAM techniques were applied to extract texture features based on remote sensing images, in order to develop an automated oil palm fruit maturity grading and sorting system. Similarly, loofah fruit changes from a smooth surface to a rough and prominent surface as it grows. Therefore, the GLCM method was used in [33] to extract texture features from loofah images to describe and quantify the relationship between pixels of different grayscale levels in the image. In

addition, the GLCM method was also effective for the maturity feature extraction of apricot in [34].

Digital image-processing technology requires preset thresholds for color and texture features in the fruit maturity stages. However, the optimal thresholds vary with the environment, especially for unstructured orchards. The influences of noise, variations in fruit surfaces, natural varying illuminations, leaf occlusions, and overlapping fruits significantly affect the parameter setting, which makes the tuning process more complicated. As a result, the accuracy and reliability of the classification model would be affected.

### 3.2. Fruit Maturity Classification

Decision-making based on image classification through machine learning algorithms is the last step in the process, which is a method of learning a set of rules from a training set to create a classifier that can be used to make a prediction using new cases in a test set. The classifier predicts the mature degree of the fruit. The most popular supervised classifiers in fruit-ripening categorization are the K-means clustering algorithm [28], SVM algorithm [30,31,36], artificial neural network (ANN), PCA algorithm [29,37], and random forest algorithm [32,33].

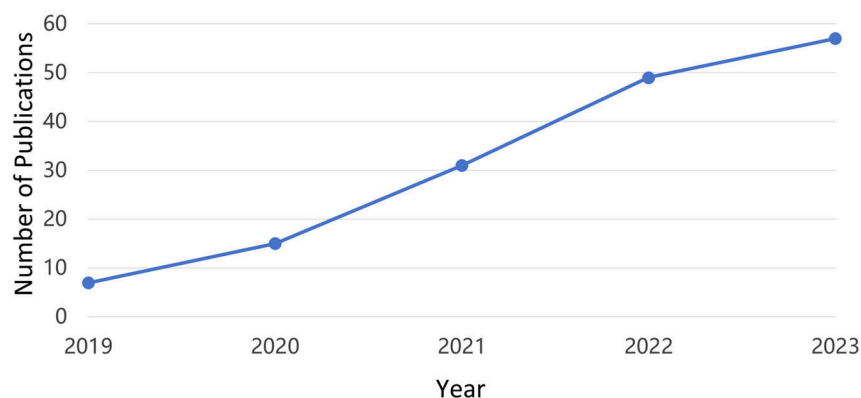
Initially, the traditional classifiers, such as K-means clustering and KNN, were developed and applied to the training set to detect the fruit region [27,28]. Li et al. [28] developed a detection method based on supervised K-means clustering for identifying blueberries in four maturity stages using natural outdoor color images, which yielded a high accuracy of 90%. The SVM algorithm is widely used in regression analysis and pattern classification. SVM aims to find an optimal hyperplane in an N-dimensional space that can classify the data points in different classes in the feature space. The hyperplane attempts to maximize the margin between the closest points of different classes. To improve the classification accuracy of fruit ripeness, Tu et al. [36] proposed a maturity grading model for passionfruit by extracting color features and using the SVM classifier, which achieved 91.52% maturity classification accuracy. Alfatni et al. compared the performance of different classifiers (SVM, KNN, etc.) for oil palm fruit maturity detection. The results showed that the SVM model was more robust than other classifiers [30]. Since machine learning is regarded as a fast, low-cost, and accurate tool for fruit grading, it has been used in mulberry-harvesting robots, where the ANN has shown a significant advantage over the SVM for mulberry ripeness classification [35]. In addition, the authors of [33] developed a random forest model, relying on the color and texture features for the three maturity classifications of the loofah fruit, with the highest accuracies of 91.42%, 90.13%, and 92.46%, respectively. However, the growth process of the loofah involves other morphological information, such as length and thickness. These features can easily be obscured in an unstructured environment, affecting the accuracy and robustness of the detection model.

Various techniques based on digital image-processing and machine learning algorithms were developed to solve the maturity detection problem for different fruits. However, the fruit grown in the natural environment would face similar color backgrounds, degrees of occlusion, fruit overlapping, and varying illumination conditions. Most of the traditional image-processing methods are sensitive to these factors, which make it difficult to extract maturity-related features and affect the detection accuracy. On the other hand, different classifiers have different limitations, for example, the SVM model requires great computation resources when dealing with a large dataset. The K-means model is sensitive to outliers and K values. Generally, the interpretability of the machine-learning-based classifiers is not sufficient.

## 4. Fruit Maturity Detection Based on Deep Learning Techniques

Deep learning has witnessed a breakthrough in computer vision and fruit detection fields. Muthulakshmi and Renjith reviewed the use of deep learning networks for fruit maturity detection [38]. Until September 2024 a total of 195 articles were found in the Web of Science database (<https://webofscience.clarivate.cn> (accessed on 19 November 2024))

using the keywords: ‘deep’ + ‘learning’ + ‘fruit’ + ‘maturity detection’. Figure 4 shows that the application of deep learning for fruit maturity detection is a popular research area with increasing growth in scientific production.



**Figure 4.** Numbers of articles published per year in the Web of Science database containing keywords ‘deep’ + ‘learning’ + ‘fruit’ + ‘maturity detection’.

The deep learning technique, represented by the convolution neural network (CNN), is a mainstream algorithm in computer vision and has been widely used in crop counting, yield estimation [39], pathogen and disease detection [40], and crop maturity classification [41]. The CNN is composed of convolutional layers, pooling layers, and fully connected layers, to process and transform the input data hierarchically. Aherwadi et al. utilized the CNN model to classify the banana maturity stages [42]. Following the idea of the CNN, many researchers were dedicated to developing various deep learning networks, such as AlexNet, VGGNet, ResNet, etc., with the aims of optimizing the network depth, accelerating the convergence rate, improving the detection accuracy, and reducing the model size. The object detection framework implements both classification and localization tasks into a single network to detect and draw the bounding box around objects in images. Depending on the network structure, the object detection network can be categorized into two-stage and one-stage detector methods.

#### 4.1. Two-Stage Detectors

The two-stage network is based on two modules: (1) a region proposal module used to propose a region of interest (RoI) containing a fruit, and (2) a classification branch adopted to classify the proposed area into the background or fruits of different ripeness and refine the detection bounding box. The most commonly used two-stage CNNs for fruit maturity detection are the Faster R-CNN [43,44] and Mask R-CNN [45], which have been adopted to detect passionfruit [36], coconut [46], tomato [47], and beef tomato [48]. The performances of these examples are summarized in Table 6.

In [46], an improved Faster R-CNN model was presented to detect tender and mature coconut in a complex environment. ResNet-50 consists of a residual module and residual connection to a deeper network to solve the problem of the vanishing gradient. The representation ability and robustness of the Faster R-CNN model can be improved by employing ResNet-50 as the backbone network. The experimental results showed that it achieved a mean average precision (mAP) of 0.894 and per image detection time of 3.124 s, and had a higher accuracy and faster speed than one-stage detectors (SSD and YOLOv3). A similar approach was used in [47], where a RoI Align module was introduced to obtain more accurate bounding boxes, and a path aggregation network was added to address the difficulty of detecting tomato maturity in complex scenarios. The mAP value and F1-score of the proposed algorithm were 96.14% and 92.95%, respectively. Compared to traditional machine learning methods, the Faster R-CNN model is implemented end-to-end with a simple feature extraction process and high accuracy. To sum up, two-stage

models are trained using well-annotated datasets, including the fruit with different stages of maturity and different growth scenarios, which further enhance the generalization and accuracy of the models. However, due to the large scale of the network, two-stage object detectors are too slow to fulfill the real-time detection needs and require more hardware storage capacity.

**Table 6.** Fruit maturity detection methods based on two-stage detectors.

| Crop        | Model                 | Accuracy     | Ref. |
|-------------|-----------------------|--------------|------|
| Fruit       | Faster R-CNN          | 98.14% (mAP) | [36] |
| Coconut     | Improved Faster R-CNN | 98.40% (mAP) | [46] |
| Tomato      | Improved Faster R-CNN | 96.14% (mAP) | [47] |
| Beef Tomato | Mask R-CNN            | 98.00% (mAP) | [48] |

#### 4.2. One-Stage Detectors

Compared with two-stage detectors, one-stage detection simultaneously predicts the object class and bounding box without using a regional proposal network. The most frequently used one-stage models include SSD [49], EfficientNet [50,51], and YOLO and its variants v3 [52], v4 [53], v5 [54], v7, and v8 [55]. The SSD was used in [56] with an image fusion algorithm to detect orange maturity. YOLOv3, YOLOv4, YOLOv5, YOLOv7, YOLOv8, and YOLOX were widely used in different fruit maturity detection works, such as blueberry, tomato, cherry tomato, strawberry, apple, mango, winter jujube, coffee, banana, citrus, and cherry, as summarized in Table 7.

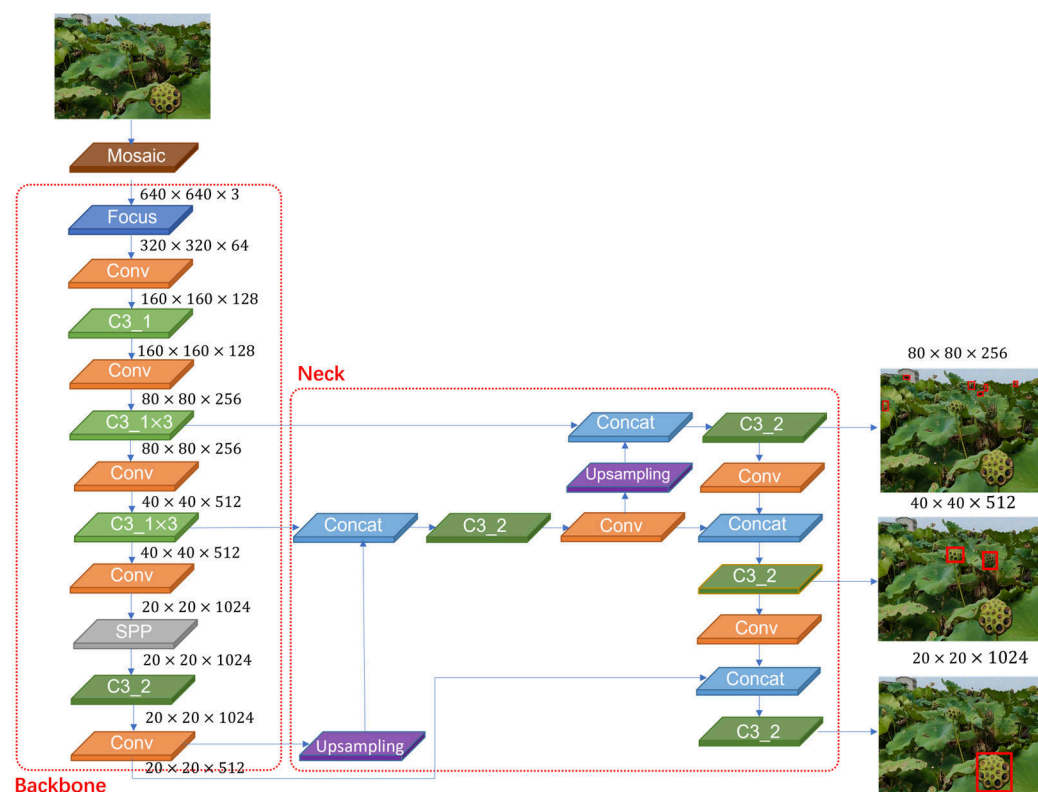
**Table 7.** Fruit maturity detection methods based on YOLO methods.

| Crop           | Model  | Improved Technique                                                     | Accuracy                             | Ref. |
|----------------|--------|------------------------------------------------------------------------|--------------------------------------|------|
| Blueberry      | YOLOv5 | I-MSRCR + CA <sup>1</sup> + GhostNet + BiFPN <sup>2</sup> + Alpha-EIoU | 98.14% (mAP)                         | [57] |
| Tomato         | YOLOv5 | EIoU                                                                   | 97.42% (mAP)                         | [58] |
| Strawberry     | YOLOX  | C3HB + NAM <sup>3</sup> + SIOU                                         | 94.26% (Precision)<br>90.7% (Recall) | [59] |
| Apple          | YOLOv5 | VoVGSCSP + GSConv + GAM <sup>4</sup>                                   | 98.7% (mAP)                          | [60] |
| Mango          | YOLOv4 | SE <sup>5</sup> + SPP <sup>6</sup>                                     | 91.91% (mAP)                         | [61] |
| Winter jujube  | YOLOv3 | Hand-engineered features                                               | 97.28% (Accuracy)                    | [62] |
| Strawberry     | YOLOv3 | None                                                                   | 89% (mAP)                            | [63] |
| Tomato         | YOLOv8 | MHSA                                                                   | 86.4% (mAP)                          | [64] |
| Cherry tomato  | YOLOv7 | Two decoders + SIOU                                                    | 84.2% (mAP)                          | [65] |
| Blueberry      | YOLOv7 | EDFM <sup>7</sup> + CARAFE <sup>8</sup>                                | 80.7% (mAP)                          | [66] |
| Coffee         | YOLOv8 | RFCACnv                                                                | 74.2% (mAP)                          | [67] |
| Banana         | YOLOv4 | None                                                                   | 99.29% (detection rate)              | [68] |
| Coffee         | YOLOv3 | None                                                                   | 84% (mAP)                            | [69] |
| Citrus         | YOLOv5 | Maximum symmetric surround saliency detection                          | 95.07% (Accuracy)                    | [70] |
| Apple          | YOLOv5 | None                                                                   | 99% (mAP)                            | [71] |
| Cherry tomato  | YOLOv8 | NanoDet detector                                                       | 90.84% (Accuracy)                    | [72] |
| Blueberry      | YOLOv5 | ShuffleNet + CBAM <sup>9</sup>                                         | 91.5% (mAP)                          | [73] |
| Apple          | YOLOv3 | DenseNet                                                               | 0.817 (F1 score)                     | [74] |
| Lotus seedpods | YOLOv8 | CBAM + WIoU                                                            | 97.8% (mAP)                          | [75] |
| Lotus seedpods | YOLOv5 | CA                                                                     | 98.3% (mAP)                          | [76] |

<sup>1</sup> CA: coordinate attention, <sup>2</sup> BiFPN: bidirectional feature pyramid network, <sup>3</sup> NAM: normalized attention module, <sup>4</sup> GAM: global attention module, <sup>5</sup> SE: squeeze and excitation, <sup>6</sup> SPP: spatial pyramid

pooling, <sup>7</sup> EDFM: enhanced detail feature module, <sup>8</sup> CARAFE: content-aware reassembly of features, and <sup>9</sup> CBAM: convolutional block attention module.

To enhance the performance of the fruit maturity detection system, various improved techniques were introduced to optimize the backbone, neck network, and prediction head of the original YOLO models. For example, to monitor the evolution of the orchard, the apple detection at several growth stages using YOLOv3 with the DenseNet backbone was presented by Tian et al., classifying young, expanding, and ripe apples with excellent results [74]. The YOLOv5 model has made great contributions to precision agriculture, and its framework is presented in Figure 5. Li et al. [58] designed a tomato maturity detection model for a harvesting system based on the YOLOv5 model, where the efficient intersection over union (EIOU) loss function was adopted to optimize the regression process of the prediction box. The proposed model yielded a precision of 95.58% and mAP value of 97.42%. A blueberry growth identification model, called YOLO-BLBE, was presented by Wang et al. In the model, the color features of blueberry in the original image can be enhanced by the I-MSRCR (Improved MSRCR (Multi-Scale Retinex with Color Restoration)) technique, where the proportion of the color restoration factors can be adjusted. Alpha-EIOU was proposed to combine three different IoU loss functions to decide the prediction box, so that the missed detection of occluded fruit could be avoided. The average identification accuracy of the model was 99.58%, 96.77%, and 98.07% for mature, semi-mature, and immature fruits, respectively [57]. Likewise, Liu et al. [60] used a depth camera to monitor the Fuji apple maturity level by employing an improved YOLOv5 model, where a lightweight convolution module and a dual-attention mechanism were introduced to reduce the parameter size and accelerate the inference speed. Estimates of the diameters of the apples were performed by applying a depth-image-based method. However, both methods were tested in a controlled laboratory setting. There are many challenges stemming from the real scenarios, such as shadows, overexposure, backlights, and background color. Furthermore, unnatural factors, including the mechanical vibration and noise, cause fruit movement, overlap, and occlusions. Consequently, researchers were dedicated to conducting ripeness detection of fruits in the real environment. Qi et al. [72] employed an advanced YOLOv8 detector to detect the contours of cherry tomatoes, which were subsequently used as inputs for ripeness prediction in a NanoDet network. This approach achieved the estimation of five maturity graded fruits with an accuracy of 90.84% and a processing rate of 15 frames per second in a structured environment. Li et al. [64] further enhanced the performance of the mature, semi-mature, and immature fruits in real environments by improving YOLOv8 with the multi-head self-attention (MHSA) mechanism. This modification led to an average accuracy of 91.6%. Nonetheless, all of the aforementioned methods detect the fruits that grow vertically downward. Regarding the fruits that grow vertically upward, Ma et al. employed an instance segmentation model based on the YOLOv8 model to carry out the detection of lotus seedpods in different poses and maturities. The average accuracies of 96% and 99% were obtained for ripe and overripe lotus seedpods, respectively. Furthermore, the proposed method effectively mitigated the errors in maturity estimation caused by occlusions, varying illuminations, and overlaps [75].



**Figure 5.** Lotus seedpods detection model based on the YOLOv5 network (image reprinted with permission from [76]).

## 5. Discussions and Future Trends

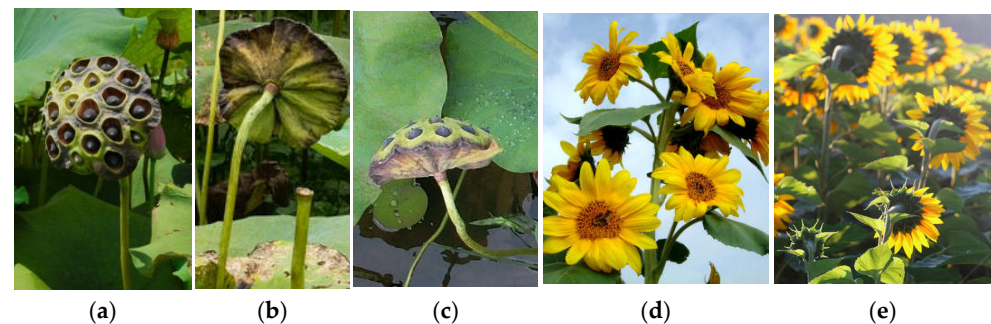
At the beginning, this review provided a summary of recent applications of non-destructive technologies on fruits maturity detection. RI, HIS, and NIR techniques possess the potential to accurately and quickly evaluate the ripening and maturity stages of fresh fruits. These approaches contain both spectral and image information that can avoid tedious and destructive sampling processes, fulfil the task of visualization, and improve the efficiency of detection. However, these techniques have not reached their full capability due to a variety of issues. For example, the main disadvantage of RI is the more expensive setup with the weaker signal than NIR. Similarly, it is required to consume more computational time for the HIS technique to acquire features from the spectral images. In terms of NIR, many factors, such as the sample characteristics, illumination condition, data pre-processing method, and calibration mode, would affect the accuracy and robustness of fruit maturity detection [8]. Recently, challenges faced by spectral imaging technology, such as higher sensitivity to illumination and atmospheric conditions and the lack of high-spatial-resolution cameras, have limited their applications in precision agriculture. In general, the increasing speed of detection algorithms and data analysis, the improvement of image-processing techniques for in-field application, and the development of low-cost hardware will attract the interest of researchers in the next stage. With the development of artificial intelligence, fruit maturity detection technology based on computer vision techniques has shown great advantages in in-field experiments.

On the basis of the literature review, the traditional digital image-processing method was first used to solve the crop detection problem in agriculture. With the wide application of artificial intelligence algorithms in computer vision, handcrafted methods have been replaced by those based on deep learning. There are various environmental factors affecting the system performance, such as lighting changes, fruit overripening, occlusions, etc., as shown in Figure 6. Nevertheless, in high-contrast background color environments, handcrafted methods showed their advantages in the low computational costs. Deep

learning has demonstrated a good performance in solving fruit maturity detection tasks in an unstructured environment. Different effective tricks were added into the backbone, neck network, or prediction head of the two-stage or one-stage models to improve model accuracy and robustness and reduce computation costs. During the literature review, we found that many works detected fruits that grow downward, and the shape of the fruits was a sphere or ellipsoid. Still, a few works focused on fruits that grow upward, and the shape of the fruits was a cone, such as lotus seedpods and sunflowers, as shown in Figure 7 [76]. Harvesting robots see fruits growing in various poses, but their maturity feature can only be seen from certain viewing angles. Thus, further efforts should be made to confront the challenges related to special-shaped fruits' ripeness detection.



**Figure 6.** Examples of fruit detection challenges under several environmental conditions: (a) fruit overlap, (b) occlusions, (c) backlight, and (d) overexposure.



**Figure 7.** Lotus seedpods and sunflowers at different viewing angles: (a) front side, (b) back side, (c) top view, (d) front side, and (e) back side.

The performance of the deep-learning-based model depends on fruit maturity datasets and the model network structure. In the aspect of the dataset, establishing and validating a detection model requires training, validation (model tuning), and test sets of images with corresponding annotation files, including the bounding box and class of the objects. In the ground truth annotation, manual bounding box annotation is tedious work and relies on the user's subjective view, especially for creating the fruit maturity dataset [39]. Due to the fact that fruit ripening is a gradual process, the plant's phenotype changes rapidly in certain growth stages, making it difficult to acquire images. Additionally, the fruits at adjacent ripening stages have no significant feature differences, which poses a challenge for image annotations. The shortage of public fruit maturity datasets creates obstacles for different detection model comparisons. On the one hand, the employment of data augmentation and synthetic data generation would be an effective way to expand the existing datasets, so that the generalization ability and robustness of the models can be improved. On the other hand, further efforts should be made to collect and amplify specialized fruit maturity datasets that contain the diversity of fruit varieties, field environments, and natural conditions. In addition, the semi-supervised learning approaches, such as pseudo-labeling and self-training techniques, have been extensively explored for their application in the fields of agriculture and manufacturing [77–79]. These techniques combine supervised and unsupervised learning approaches, utilizing a small amount of labeled data to train a pretrained model and employing a larger amount of unlabeled data to enhance the model's generalization capability. They can effectively address the

challenges of fruit maturity image acquisition, making it easy to achieve high-performance detection.

As summarized and reviewed in this paper, the detection model network based on the YOLO series model achieved a better accuracy rate in addressing environmental challenges compared with other approaches. Furthermore, many tricks, such as the attention mechanism [80–82], Swin transformer [83], multi-feature fusion [84], improved loss function [85], and data enhancement approaches [86], should be paid more attention to strengthen the deep learning models' performance. Further research will emphasize improving the accuracy, speed, robustness, and generalization of vision systems, while reducing the overall complexity and costs. Developing optimization strategies for deep learning networks plays vital roles in saving the computation resources and improving the inference efficiency. For example, the application of model pruning and knowledge distillation reduces the parameters and complexity of the model while maintaining the model accuracy. Having information on the growth stage or maturity stage of fruits and their location in the environment will allow farms to schedule harvesting seasons and transportation times. More informed agronomic decisions can be made within the framework of precision agriculture to improve fruit quality and maximize profits.

## 6. Conclusions

From the results of the literature review on fruit maturity detection, it can be summarized that although remarkable developments have been achieved in recent years, it remains the focus of current research. Fruit harvesting and yield assessment in orchards or greenhouses is usually carried out manually, which is limited by the shortage of labor. Developing intelligent fruit harvesting and grading systems is an efficient way to expand the scale of the fruit industry.

The emerging techniques, including spectral analysis, traditional digital image-processing, machine learning, and deep learning models, have a wide application prospect in fruit maturity detection. Regarding the spectral analysis technique, the advantages and disadvantages of the RI, NIR, and HIS techniques were summarized and analyzed. The HIS technique is restricted by high costs and long processing times. Developing a low-cost real-time HIS technique to improve the processing speed will be a future trend. Meanwhile, various image-processing methods with machine-learning-based classifiers were presented. The computation costs of these methods are relatively low, but the model performance is greatly affected by environmental conditions. These obstacles were solved by deep learning models. The application of a one-stage detector effectively improved the accuracy, efficiency, and robustness of the fruit detection. The detection performance can be further improved by strengthening the dataset and model. Therefore, developing a data augmentation method to create a large dataset or introducing the semi-supervised learning method will be future research directions.

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